



DC – AI – Routing – Switching – Security – Wireless – Automation

ACX7100/ACX7509 Validation: L3VPN/6VPE

An HPE Juniper Networking Techpost

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HPE



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1 L3VPN Validation on ACX7000

Suneesh Babu - 11/09/2022

ACX7000 platform has been tested successfully with 4,000 Layer3 VPN Routing-instances with BGPv4, BGPv6, OSPF, OSPFv3, ISISv4, ISISv6, Static-v4, Static-v6 as CE-PE protocols and with a total of 1.35M routes.

1.1 Introduction

This is the third article in the ACX7k Metro Validation Series.

In this article we validate the L3VPN, 6VPE, 6PE feature scale support on ACX7100-32C platform with 22.2R1 Junos-EVO build.

[RFC 4364](#) describes BGP-enabled [L3VPN](#) to connect Customer Edge Devices (CEs) across a Provider Edge (PE) Network. The PEs provide layer 3 connectivity for their customers using various protocols like [BGP](#), [OSPF](#), [ISIS](#), [RIP](#) or [Static](#). The PEs of the network are connected via MPLS technology [LDP](#), [RSVP](#), [SR-MPLS](#) etc. In L3VPN, the IP address associated with the hosts are learned via the control plane on both CE-PE side and across PE network with the help of BGP.

JUNOS-EVO supports L3VPN and 6VPE functionality with instance-type “vrf”. In this document, we qualify the feature scale on the ACX7000 platforms.

[ACX7100-32C](#) with 22.2R1 Junos Version supports different flavours of CE-PE. The Key Performance Indicators (KPI) tested are captured in the below table. The same scale is supported and tested on [ACX7100-48L](#) as well as on [ACX7509](#). But the [ACX7024](#) Scale is not covered in this article (same features but different scale are expected on this platform).

Please note that despite sharing the same ACX moniker, the ACX7000 products are different products than ACX500/710/1000/1100/2100/2200/4000/5000/5400/6000. They are powered by different Packet Forwarding Engines (PFE), and support different feature sets and scales.

KPI	Scale
L3VPN Instances	4,000
6VPE Instances (Dual Stack)	1,150
BGPv4 Peering (CE-PE)	2,000
BGPv6 Peering (CE-PE)	500
OSPF Neighbors (CE-PE)	1,000
OSPFv3 Neighbors (CE-PE)	100
ISIS IPv4 Adjacencies (CE-PE)	500
ISIS IPv6 Adjacencies (CE-PE)	500
Static-v4 Peering (CE-PE)	500
Static-v6 Peering (CE-PE)	50
VPNv4 Routes	1,151,000

KPI	Scale
VPNv6 Routes	115,000
IPv4 Global Routes	60,000
IPv6 Global Routes (6PE)	30,000
Total Route in the DUT	1,350,000

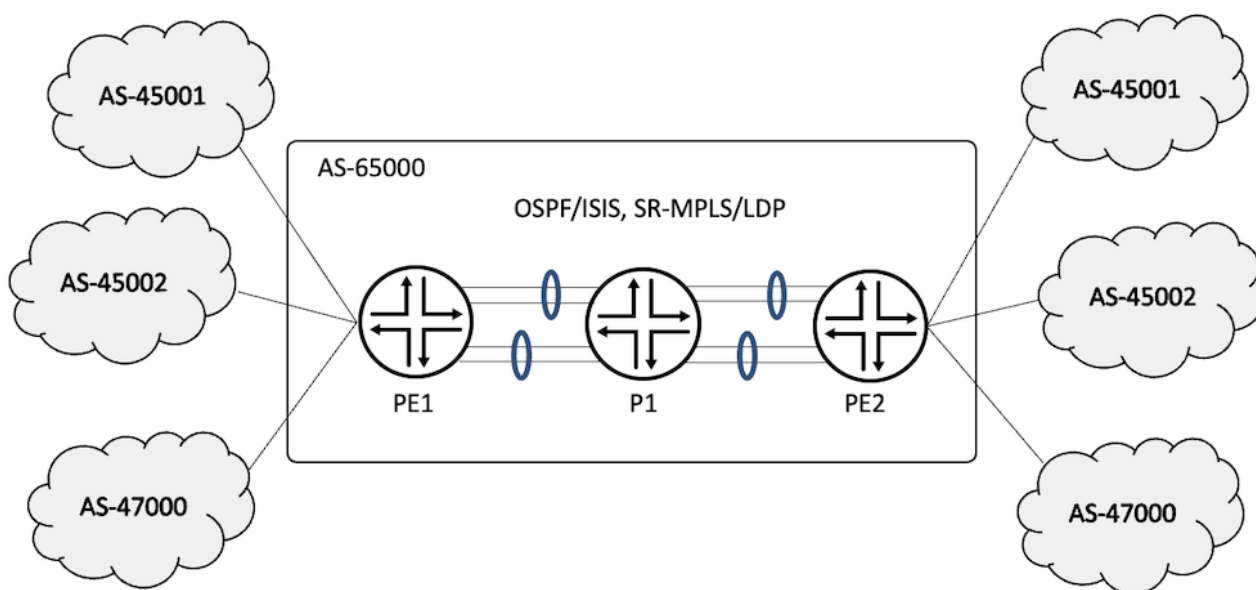
1.2 Metro L3VPN Topology

Test topology consists of three routers:

- an ACX7100-32C is the PE1
- an ACX7100-48L is the PE2
- a [PTX10008](#) is the P1.

CEs are simulated using a traffic generator. PEs are connected to CEs using 5x 100G links and to the core using 8x 100G links, which are bundled in 2x LAGs each having four links.

The ACX7100-32C is the Device Under Test (DUT). The underlay transport used for testing is SR-MPLS with both OSPF and ISIS protocols individually. We also covered LDP as the underlay transport. The CE-bound interface configurations are of SP-Style (see glossary).



For this test, we configure 4,000 L3VPN instances:

- 2,000 have EBGPv4 as the CE-PE protocol,
- 1,000 have OSPF as the CE-PE protocol,
- 500 have ISIS as the CE-PE protocol
- the remaining 500 have Static-v4 reachability.
- 1,150 routing instances are
 - dual-stacked,

- with 500 EBGPv6 Peering,
 - 100 OSPFv3 Peering,
 - 500 IS-IS Peering
 - 50 Static IPv6 reachability.
- The DUT learns 1.35M routes including VPNv4, VPNv6, global IPv4 and IPv6.

Bidirectional traffic in iMIX mode at 99.9% offer-load flows for all the L3VPN, 6VPE services. A total of ~1Tbps traffic transits through the DUT.

1.2.1 Base Configuration

The IGP and BGP configuration required to bring up L3VPN, 6VPE, 6PE are as follows:

```

1 regress@PE1> show configuration protocols |display inheritance no-comments
2 bgp {
3   group IBGP {
4     type internal;
5     local-address 12.1.1.1;
6     family inet {
7       unicast;
8     }
9     family inet-vpn {
10      unicast;
11    }
12    family inet6 {
13      labeled-unicast;
14    }
15    family inet6-vpn {
16      unicast;
17    }
18    neighbor 12.1.1.3 {
19      export nhself;
20    }
21  }
22  group EBG {
23    type external;
24    as-override;
25    neighbor 16.100.1.2 {
26      family inet {
27        unicast;
28      }
29      peer-as 65001;
30    }
31    neighbor 2002:16:100:1::2 {
32      family inet6 {
33        unicast;
34      }
35      peer-as 65001;
36    }
37  }
38 }
39 mpls {
40   icmp-tunneling;
41   ipv6-tunneling;
42   interface ae0.0;
43   interface ae1.0;
44 }
45 ospf {
46   source-packet-routing {
47     node-segment ipv4-index 101;
48     srgb start-label 45000 index-range 2000;
49   }
50   area 0.0.0.0 {
51     interface lo0.0 {
52       passive;
53     }
54     interface ae0.0 {
55       interface-type p2p;
56     }
57     interface ae1.0 {
58       interface-type p2p;

```

```

59     }
60   }
61 }
62
63 regress@PE1>

```

The following configurations brings-up BGPv4 and BGPv6 Peering on a routing-instance.

```

1  interfaces {
2    et-0/0/0 {
3      description To_VPNv4_VPNv6_CE;
4      flexible-vlan-tagging;
5      mtu 9022;
6      encapsulation flexible-ethernet-services;
7      unit 1 {
8        vlan-id 1;
9        family inet {
10         address 16.0.1.1/24;
11       }
12       family inet6 {
13         address 2002:16:0:1::1/64;
14       }
15     }
16   }
17 }
18 routing-instances {
19   METRO_VRF_1 {
20     instance-type vrf;
21     protocols {
22       bgp {
23         group METRO_VRF_EBGPv4_1 {
24           type external;
25           family inet {
26             unicast;
27           }
28           peer-as 45001;
29           as-override;
30           neighbor 16.0.1.2;
31         }
32         group METRO_VRF_EBGPv6_1 {
33           type external;
34           family inet6 {
35             unicast;
36           }
37           peer-as 45001;
38           as-override;
39           neighbor 2002:16:0:1::2;
40         }
41       }
42     }
43     interface et-0/0/0.1;
44     route-distinguisher 12.1.1.1:1;
45     vrf-target target:65000:1;
46     vrf-table-label;
47   }
48 }

```

The following configuration brings up OSPF and OSPFv3 for a routing-instance

```

1  interfaces {
2    et-0/0/2 {
3      description To_VPNv4_OSPF_CE;
4      flexible-vlan-tagging;
5      mtu 9022;
6      encapsulation flexible-ethernet-services;
7      unit 1 {
8        vlan-id 1;
9        family inet {
10         address 16.10.1.1/24;
11       }
12       family inet6 {
13         address 2002:16:10:1::1/64;
14       }
15     }
16   }
17 }
18 routing-instances {
19   METRO_VRF_2001 {
20     instance-type vrf;

```

```

21     protocols {
22         ospf {
23             area 0.0.0.0 {
24                 interface et-0/0/2.1 {
25                     interface-type p2p;
26                 }
27             }
28         }
29         ospf3 {
30             area 0.0.0.0 {
31                 interface et-0/0/2.1 {
32                     interface-type p2p;
33                 }
34             }
35         }
36     }
37     interface et-0/0/2.1;
38     route-distinguisher 12.1.1.1:2001;
39     vrf-target target:65000:2001;
40     vrf-table-label;
41 }
42 }

```

Configuration to bring up ISISv4 and ISISv6 on a routing-instance:

```

1  interfaces {
2      et-0/0/3 {
3          description To_VPNv4_ISIS_CE;
4          flexible-vlan-tagging;
5          mtu 9022;
6          encapsulation flexible-ethernet-services;
7          unit 1 {
8              vlan-id 1;
9              family inet {
10                 address 16.14.1.1/24;
11             }
12             family iso;
13             family inet6 {
14                 address 2002:16:14:1::1/64;
15             }
16         }
17     }
18     lo0 {
19         unit 1 {
20             family inet {
21                 address 12.2.0.1/32;
22             }
23             family iso {
24                 address 49.0001.0012.0002.0000.0001.00;
25             }
26             family inet6 {
27                 address 2005:12:2::1/128;
28             }
29         }
30     }
31 }
32 routing-instances {
33     METRO_VRF_3001 {
34         instance-type vrf;
35         protocols {
36             isis {
37                 interface et-0/0/3.1 {
38                     level 1 disable;
39                     point-to-point;
40                 }
41                 interface lo0.1 {
42                     passive;
43                 }
44             }
45         }
46         interface et-0/0/3.1;
47         interface lo0.1;
48         route-distinguisher 12.1.1.1:3001;
49         vrf-target target:65000:3001;
50         vrf-table-label;
51     }
52 }

```

Configuration to bring up Staticv4 and Staticv6 for a routing-instance:

```

1  interfaces {
2      et-0/0/4 {
3          description To_VPNv4_Static_CE;
4          flexible-vlan-tagging;
5          mtu 9022;
6          encapsulation flexible-ethernet-services;
7          unit 1 {
8              vlan-id 1;
9              family inet {
10                 address 16.16.1.1/24;
11             }
12             family inet6 {
13                 address 2002:16:16:1::1/64;
14             }
15         }
16     }
17 }
18 routing-instances {
19     METRO_VRF_3501 {
20         instance-type vrf;
21         routing-options {
22             rib METRO_VRF_3501.inet6.0 {
23                 static {
24                     route 2008:1:0:1::/64 next-hop 2002:16:16:1::2;
25                 }
26             }
27             static {
28                 route 46.0.1.0/24 next-hop 16.16.1.2;
29             }
30         }
31         interface et-0/0/4.1;
32         route-distinguisher 12.1.1.1:3501;
33         vrf-target target:65000:3501;
34         vrf-table-label;
35     }
36 }

```

1.2.2 VPN CE Bound Interface Schema

The five CE bound 100G interfaces are logically split across the routing instances and detailed in this chart:

Metro L3VPN, 6VPE, 6PE and Global IPv4

Port

P-1

EBGPv6

P-2

Global v4

Global v6

P-3

OSPFv3

P-4

ISISv6

P-5

Staticv6

1.3 L3VPN Verification

The first step to verify the L3VPN service is to make sure that the IGP is established and routers in the core network are reachable. BGP Peering has to be verified for various services like L3VPN, 6VPE, 6PE and Global V4 routes. The L3VPN routes are present in bgp.l3vpn.0 table, 6VPE routes in bgp.l3vpn-inet6.0 table, 6PE routes in inet6.0 table and Global IPv4 routes in inet.0 table.

1.3.1 Core IGP Verification

The neighborship in the core router and their reachability are verified as follows:

```

1 regress@PE1> show ospf neighbor
2 Address Interface State ID Pri Dead
3 15.1.1.2 ae0.0 Full 6.6.6.6 128 37
4 15.1.2.2 ae1.0 Full 6.6.6.6 128 31
5 regress@PE1> show route protocol ospf table inet.0
6
7 inet.0: 60031 destinations, 60038 routes (60031 active, 0 holddown, 0 hidden)
8 + = Active Route, - = Last Active, * = Both
9
10 12.1.1.2/32 *[OSPF/10/10] 1d 17:40:52, metric 1
11      to 15.1.1.2 via ae0.0
12      > to 15.1.2.2 via ae1.0
13 12.1.1.3/32 *[OSPF/10/10] 1d 17:40:52, metric 2
14      to 15.1.1.2 via ae0.0
15      > to 15.1.2.2 via ae1.0
16 15.2.1.0/24 *[OSPF/10/10] 1d 17:40:52, metric 2
17      to 15.1.1.2 via ae0.0
18      > to 15.1.2.2 via ae1.0
19 15.2.2.0/24 *[OSPF/10/10] 1d 17:40:52, metric 2
20      to 15.1.1.2 via ae0.0
21      > to 15.1.2.2 via ae1.0
22 224.0.0.5/32 *[OSPF/10] 1d 17:41:08, metric 1
23      MultiRecv
24 224.0.0.6/32 *[OSPF/10] 1d 17:41:08, metric 1
25      MultiRecv
26
27 regress@PE1> show route protocol ospf table inet.3
28
29 inet.3: 2 destinations, 2 routes (2 active, 0 holddown, 0 hidden)
30 + = Active Route, - = Last Active, * = Both
31
32 12.1.1.2/32 *[L-OSPF/10/5] 00:13:20, metric 1
33      to 15.1.1.2 via ae0.0
34      > to 15.1.2.2 via ae1.0
35 12.1.1.3/32 *[L-OSPF/10/5] 1d 17:40:55, metric 2
36      to 15.1.1.2 via ae0.0, Push 45103
37      > to 15.1.2.2 via ae1.0, Push 45103
38
39 regress@PE1> show mpls interface
40 Interface State Administrative groups (x: extended)
41 ae0.0 Up <none>
42 ae1.0 Up <none>
43
44 regress@PE1>

```

1.3.2 Core BGP Verification

```

1 regress@PE1> show bgp summary
2 Threading mode: BGP I/O
3 Default eBGP mode: advertise - accept, receive - accept
4 Groups: 2502 Peers: 2503 Down peers: 0
5 Table Tot Paths Act Paths Suppressed History Damp State Pending
6 inet.0
7      60009 60002 0 0 0 0
8 inet6.0
9      30002 30002 0 0 0 0
10 bgp.l3vpn.0
11      580000 580000 0 0 0 0

```

```

12 bgp.l3vpn-inet6.0
13      61700 61700 0 0 0 0
14 inet6.3
15      0 0 0 0 0 0
16 Peer AS InPkt OutPkt OutQ Flaps Last Up/Dwn State|#Active/Received/Accepted/Damped...
17 12.1.1.3 65000 21313 20562 0 0 1d 17:40:57 Establ
18   inet.0: 30002/30009/30009/0
19   inet6.0: 15002/15002/15002/0
20   bgp.l3vpn.0: 580000/580000/580000/0
21   bgp.l3vpn-inet6.0: 61700/61700/61700/0
22   METRO_VRF_1.inet.0: 251/251/251/0
23   METRO_VRF_1.inet6.0: 101/101/101/0
24   METRO_VRF_2.inet.0: 251/251/251/0
25   METRO_VRF_2.inet6.0: 101/101/101/0
26   METRO_VRF_3.inet.0: 251/251/251/0
27   METRO_VRF_3.inet6.0: 101/101/101/0
28   ...
29   METRO_VRF_1997.inet.0: 251/251/251/0
30   METRO_VRF_1998.inet.0: 251/251/251/0
31   METRO_VRF_1999.inet.0: 251/251/251/0
32   METRO_VRF_2000.inet.0: 251/251/251/0
33   METRO_VRF_2001.inet.0: 51/51/51/0
34   METRO_VRF_2002.inet.0: 51/51/51/0
35   METRO_VRF_2003.inet.0: 51/51/51/0
36   ...
37   METRO_VRF_2998.inet.0: 51/51/51/0
38   METRO_VRF_2999.inet.0: 51/51/51/0
39   METRO_VRF_3000.inet.0: 51/51/51/0
40   METRO_VRF_3001.inet.0: 52/52/52/0
41   METRO_VRF_3002.inet.0: 52/52/52/0
42   METRO_VRF_3003.inet.0: 52/52/52/0
43   ...
44   METRO_VRF_3498.inet.0: 52/52/52/0
45   METRO_VRF_3499.inet.0: 52/52/52/0
46   METRO_VRF_3500.inet.0: 52/52/52/0
47   METRO_VRF_3501.inet.0: 2/2/2/0
48   METRO_VRF_3502.inet.0: 2/2/2/0
49   METRO_VRF_3503.inet.0: 2/2/2/0
50   ...
51   METRO_VRF_3998.inet.0: 2/2/2/0
52   METRO_VRF_3999.inet.0: 2/2/2/0
53   METRO_VRF_4000.inet.0: 2/2/2/0
54   METRO_VRF_2001.inet6.0: 51/51/51/0
55   METRO_VRF_2002.inet6.0: 51/51/51/0
56   METRO_VRF_2003.inet6.0: 51/51/51/0
57   ...
58   METRO_VRF_2098.inet6.0: 51/51/51/0
59   METRO_VRF_2099.inet6.0: 51/51/51/0
60   METRO_VRF_2100.inet6.0: 51/51/51/0
61   METRO_VRF_3001.inet6.0: 12/12/12/0
62   METRO_VRF_3002.inet6.0: 12/12/12/0
63   METRO_VRF_3003.inet6.0: 12/12/12/0
64   ...
65   METRO_VRF_3548.inet6.0: 2/2/2/0
66   METRO_VRF_3549.inet6.0: 2/2/2/0
67   METRO_VRF_3550.inet6.0: 2/2/2/0
68 16.0.1.2 45001 5003 5046 0 0 1d 17:39:49 Establ
69   METRO_VRF_1.inet.0: 250/250/250/0
70 16.0.2.2 45002 5003 5046 0 0 1d 17:39:49 Establ
71   METRO_VRF_2.inet.0: 250/250/250/0
72 16.0.3.2 45003 5003 5046 0 0 1d 17:39:54 Establ
73   METRO_VRF_3.inet.0: 250/250/250/0
74   ...
75 2002:16:0:1f3::2 45499 5003 5046 0 0 1d 17:39:32 Establ
76   METRO_VRF_499.inet6.0: 100/100/100/0
77 2002:16:0:1f4::2 45500 5003 5046 0 0 1d 17:39:32 Establ
78   METRO_VRF_500.inet6.0: 100/100/100/0
79 2002:16:100:1::2 65001 5036 5080 0 0 1d 17:39:59 Establ
80   inet6.0: 15000/15000/15000/0
81
82 regress@PE1>

```

1.3.3 BGPv4 CE-PE Verification

The BGPv4 neighborhood in one of the routing-instances METRO_VRF_1 is as follows:

```

1 regress@PE1> show bgp neighbor 16.0.1.2
2 Peer: 16.0.1.2+33773 AS 45001 Local: 16.0.1.1+179 AS 65000
3 Group: METRO_VRF_EBGPv4_1 Routing-Instance: METRO_VRF_1
4 Forwarding routing-instance: METRO_VRF_1
5 Type: External State: Established Flags: <Sync>
6 Last State: OpenConfirm Last Event: RecvKeepAlive
7 Last Error: None
8 Options: <AddressFamily PeerAS Refresh As Override>
9 Options: <GracefulShutdownRcv>
10 Address families configured: inet-unicast
11 Holdtime: 90 Preference: 170
12 Graceful Shutdown Receiver local-preference: 0
13 Number of flaps: 0
14 Receive eBGP Origin Validation community: Reject
15 Peer ID: 192.0.0.1 Local ID: 16.0.1.1 Active Holdtime: 90
16 Keepalive Interval: 30 Group index: 2 Peer index: 0 SNMP index: 3
17 I/O Session Thread: bgpio-0 State: Enabled
18 BFD: disabled, down
19 Local Interface: et-0/0/0.1
20 NLRI for restart configured on peer: inet-unicast
21 NLRI advertised by peer: inet-unicast inet-multicast inet-vpn-unicast inet6-unicast inet6-
   multicast l2vpn inet6-vpn-unicast
22 NLRI for this session: inet-unicast
23 Peer supports Refresh capability (2)
24 Stale routes from peer are kept for: 300
25 Peer does not support Restarter functionality
26 NLRI of all end-of-rib markers sent: inet-unicast
27 Peer does not support Receiver functionality
28 Peer does not support LLGR Restarter or Receiver functionality
29 Peer does not support 4 byte AS extension
30 Peer does not support Addpath
31 Table METRO_VRF_1.inet.0 Bit: 60001
32 RIB State: BGP restart is complete
33 RIB State: VPN restart is complete
34 Send state: in sync
35 Active prefixes: 250
36 Received prefixes: 250
37 Accepted prefixes: 250
38 Suppressed due to damping: 0
39 Advertised prefixes: 251
40 Last traffic (seconds): Received 1 Sent 18 Checked 154592
41 Input messages: Total 5157 Updates 1 Refreshes 0 Octets 99082
42 Output messages: Total 5201 Updates 2 Refreshes 0 Octets 99895
43 Output Queue[5]: 0 (METRO_VRF_1.inet.0, inet-unicast)
44
45 regress@PE1>

```

The BGPv4 CE-PE routes in the routing-instance METRO_VRF_1 are as follows:

```

1 regress@PE1> show route table METRO_VRF_1.inet.0
2
3 METRO_VRF_1.inet.0: 503 destinations, 503 routes (503 active, 0 holddown, 0 hidden)
4 + = Active Route, - = Last Active, * = Both
5
6 16.0.1.0/24 *[Direct/0] 1d 18:03:51
7   > via et-0/0/0.1
8 16.0.1.1/32 *[Local/0] 1d 18:03:51
9   Local via et-0/0/0.1
10 17.0.1.0/24 *[BGP/170] 1d 17:53:50, localpref 100, from 12.1.1.3
11   AS path: I, validation-state: unverified
12   > to 15.1.1.2 via ae0.0, Push 16, Push 45103(top)
13   to 15.1.2.2 via ae1.0, Push 16, Push 45103(top)
14 20.0.1.0/24 *[BGP/170] 1d 17:52:42, localpref 100
15   AS path: 45001 I, validation-state: unverified
16   > to 16.0.1.2 via et-0/0/0.1
17 20.0.2.0/24 *[BGP/170] 1d 17:52:42, localpref 100
18   AS path: 45001 I, validation-state: unverified
19   > to 16.0.1.2 via et-0/0/0.1
20 ...
21 20.0.249.0/24 *[BGP/170] 1d 17:52:42, localpref 100
22   AS path: 45001 I, validation-state: unverified
23   > to 16.0.1.2 via et-0/0/0.1
24 20.0.250.0/24 *[BGP/170] 1d 17:52:42, localpref 100
25   AS path: 45001 I, validation-state: unverified
26   > to 16.0.1.2 via et-0/0/0.1
27 30.0.1.0/24 *[BGP/170] 1d 17:52:39, localpref 100, from 12.1.1.3
28   AS path: 45001 I, validation-state: unverified
29   to 15.1.1.2 via ae0.0, Push 16, Push 45103(top)
30   > to 15.1.2.2 via ae1.0, Push 16, Push 45103(top)
31 30.0.2.0/24 *[BGP/170] 1d 17:52:39, localpref 100, from 12.1.1.3

```

```

32         AS path: 45001 I, validation-state: unverified
33         to 15.1.1.2 via ae0.0, Push 16, Push 45103(top)
34         > to 15.1.2.2 via ae1.0, Push 16, Push 45103(top)
35     ...
36 30.0.249.0/24 *[BGP/170] 1d 17:52:39, localpref 100, from 12.1.1.3
37     AS path: 45001 I, validation-state: unverified
38     > to 15.1.1.2 via ae0.0, Push 16, Push 45103(top)
39     to 15.1.2.2 via ae1.0, Push 16, Push 45103(top)
40 30.0.250.0/24 *[BGP/170] 1d 17:52:39, localpref 100, from 12.1.1.3
41     AS path: 45001 I, validation-state: unverified
42     > to 15.1.1.2 via ae0.0, Push 16, Push 45103(top)
43     to 15.1.2.2 via ae1.0, Push 16, Push 45103(top)
44
45 regress@PE1>

```

1.3.4 BGPv6 CE-PE Verification

The BGPv6 neighborship in one of the routing-instance METRO_VRF_1:

```

1 regress@PE1> show bgp neighbor 2002:16:0:1::2
2 Peer: 2002:16:0:1::2+32777 AS 45001 Local: 2002:16:0:1::1+179 AS 65000
3   Group: METRO_VRF_EBGPv6_1 Routing-Instance: METRO_VRF_1
4   Forwarding routing-instance: METRO_VRF_1
5   Type: External State: Established Flags: <Sync>
6   Last State: OpenConfirm Last Event: RecvKeepAlive
7   Last Error: None
8   Options: <AddressFamily PeerAS Refresh As Override>
9   Options: <GracefulShutdownRcv>
10  Address families configured: inet6-unicast
11  Holdtime: 90 Preference: 170
12  Graceful Shutdown Receiver local-preference: 0
13  Number of flaps: 0
14  Receive eBGP Origin Validation community: Reject
15  Peer ID: 193.0.0.1 Local ID: 16.0.1.1 Active Holdtime: 90
16  Keepalive Interval: 30 Group index: 3 Peer index: 0 SNMP index: 4
17  I/O Session Thread: bgpio-0 State: Enabled
18  BFD: disabled, down
19  Local Interface: et-0/0/0.1
20  NLRI for restart configured on peer: inet6-unicast
21  NLRI advertised by peer: inet-unicast inet-multicast inet-vpn-unicast inet6-
   multicast l2vpn inet6-vpn-unicast
22  NLRI for this session: inet6-unicast
23  Peer supports Refresh capability (2)
24  Stale routes from peer are kept for: 300
25  Peer does not support Restarter functionality
26  NLRI of all end-of-rib markers sent: inet6-unicast
27  Peer does not support Receiver functionality
28  Peer does not support LLGR Restarter or Receiver functionality
29  Peer does not support 4 byte AS extension
30  Peer does not support Addpath
31  NLRI that peer supports extended nexthop encoding for: inet-unicast inet-multicast inet-vpn-
   unicast
32  Table METRO_VRF_1.inet6.0 Bit: 70001
33  RIB State: BGP restart is complete
34  RIB State: VPN restart is complete
35  Send state: in sync
36  Active prefixes: 100
37  Received prefixes: 100
38  Accepted prefixes: 100
39  Suppressed due to damping: 0
40  Advertised prefixes: 101
41  Last traffic (seconds): Received 29 Sent 24 Checked 154980
42  Input messages: Total 5169 Updates 1 Refreshes 0 Octets 99266
43  Output messages: Total 5213 Updates 2 Refreshes 0 Octets 100103
44  Output Queue[6]: 0 (METRO_VRF_1.inet6.0, inet6-unicast)
45
46 regress@PE1>

```

The BGPv6 CE-PE routes in the routing-instance METRO_VRF_1:

```

1 regress@PE1> show route table METRO_VRF_1.inet6.0
2
3 METRO_VRF_1.inet6.0: 205 destinations, 205 routes (205 active, 0 holddown, 0 hidden)
4 + = Active Route, - = Last Active, * = Both
5

```

```

6 2002:16:0:1::/64 *[Direct/0] 1d 18:07:23
7   > via et-0/0/0.1
8 2002:16:0:1::1/128 *[Local/0] 1d 18:07:23
9   Local via et-0/0/0.1
10 2002:17:0:1::/64 *[BGP/170] 1d 17:57:32, localpref 100, from 12.1.1.3
11   AS path: I, validation-state: unverified
12   > to 15.1.1.2 via ae0.0, Push 16, Push 2, Push 45103(top)
13   to 15.1.2.2 via ae1.0, Push 16, Push 2, Push 45103(top)
14 2002:20:0:1::/64 *[BGP/170] 1d 17:56:28, localpref 100
15   AS path: 45001 I, validation-state: unverified
16   > to 2002:16:0:1::2 via et-0/0/0.1
17 2002:20:0:2::/64 *[BGP/170] 1d 17:56:28, localpref 100
18   AS path: 45001 I, validation-state: unverified
19   > to 2002:16:0:1::2 via et-0/0/0.1
20 ...
21 2002:20:0:63::/64 *[BGP/170] 1d 17:56:28, localpref 100
22   AS path: 45001 I, validation-state: unverified
23   > to 2002:16:0:1::2 via et-0/0/0.1
24 2002:20:0:64::/64 *[BGP/170] 1d 17:56:28, localpref 100
25   AS path: 45001 I, validation-state: unverified
26   > to 2002:16:0:1::2 via et-0/0/0.1
27 2002:30:0:1::/64 *[BGP/170] 1d 17:56:27, localpref 100, from 12.1.1.3
28   AS path: 45001 I, validation-state: unverified
29   > to 15.1.1.2 via ae0.0, Push 16, Push 2, Push 45103(top)
30   to 15.1.2.2 via ae1.0, Push 16, Push 2, Push 45103(top)
31 2002:30:0:2::/64 *[BGP/170] 1d 17:56:27, localpref 100, from 12.1.1.3
32   AS path: 45001 I, validation-state: unverified
33   > to 15.1.1.2 via ae0.0, Push 16, Push 2, Push 45103(top)
34   to 15.1.2.2 via ae1.0, Push 16, Push 2, Push 45103(top)
35 ...
36 2002:30:0:63::/64 *[BGP/170] 1d 17:56:27, localpref 100, from 12.1.1.3
37   AS path: 45001 I, validation-state: unverified
38   > to 15.1.1.2 via ae0.0, Push 16, Push 2, Push 45103(top)
39   to 15.1.2.2 via ae1.0, Push 16, Push 2, Push 45103(top)
40 2002:30:0:64::/64 *[BGP/170] 1d 17:56:27, localpref 100, from 12.1.1.3
41   AS path: 45001 I, validation-state: unverified
42   > to 15.1.1.2 via ae0.0, Push 16, Push 2, Push 45103(top)
43   to 15.1.2.2 via ae1.0, Push 16, Push 2, Push 45103(top)
44 fe80::ae78:d100:12d:3504/128
45   *[Local/0] 1d 18:07:23
46   Local via et-0/0/0.1
47 ff02::2/128 *[INET6/0] 1d 18:10:03
48   MultiRecv
49
50 regress@PE1>

```

1.3.5 CE-PE OSPF Verification

OSPF Neighborhood of the routing-instance is as shown below:

```

1 regress@PE1> show ospf neighbor instance METRO_VRF_2001
2 Address Interface State ID Pri Dead
3 16.10.1.2 et-0/0/2.1 Full 195.0.0.1 0 30

```

The OSPF routes received from the CE can be listed as:

```

1 regress@PE1> show route table METRO_VRF_2001.inet.0 protocol ospf
2
3 METRO_VRF_2001.inet.0: 105 destinations, 105 routes (105 active, 0 holddown, 0 hidden)
4 + = Active Route, - = Last Active, * = Both
5
6 40.0.1.0/24 *[OSPF/10] 1d 16:50:57, metric 1
7   > to 16.10.1.2 via et-0/0/2.1
8 40.0.2.0/24 *[OSPF/10] 1d 16:50:57, metric 1
9   > to 16.10.1.2 via et-0/0/2.1
10 ...
11 40.0.49.0/24 *[OSPF/10] 1d 16:50:57, metric 1
12   > to 16.10.1.2 via et-0/0/2.1
13 40.0.50.0/24 *[OSPF/10] 1d 16:50:57, metric 1
14   > to 16.10.1.2 via et-0/0/2.1
15 224.0.0.5/32 *[OSPF/10] 1d 18:13:21, metric 1
16   MultiRecv
17 224.0.0.6/32 *[OSPF/10] 1d 18:13:21, metric 1
18   MultiRecv
19

```

```
20 regress@PE1>
```

The OSPF received from the PE2 routes as BGP routes over the L3VPN BGP family can be listed as follows.

```
1 regress@PE1> show route table METRO_VRF_2001.inet.0 protocol bgp
2
3 METRO_VRF_2001.inet.0: 105 destinations, 105 routes (105 active, 0 holddown, 0 hidden)
4 + = Active Route, - = Last Active, * = Both
5
6 17.10.1.0/24 *[BGP/170] 1d 18:02:09, localpref 100, from 12.1.1.3
7     AS path: I, validation-state: unverified
8     > to 15.1.1.2 via ae0.0, Push 1131, Push 45103(top)
9     > to 15.1.2.2 via ae1.0, Push 1131, Push 45103(top)
10 42.0.1.0/24 *[BGP/170] 1d 17:24:52, MED 1, localpref 100, from 12.1.1.3
11     AS path: I, validation-state: unverified
12     > to 15.1.1.2 via ae0.0, Push 1131, Push 45103(top)
13     > to 15.1.2.2 via ae1.0, Push 1131, Push 45103(top)
14 42.0.2.0/24 *[BGP/170] 1d 17:24:52, MED 1, localpref 100, from 12.1.1.3
15     AS path: I, validation-state: unverified
16     > to 15.1.1.2 via ae0.0, Push 1131, Push 45103(top)
17     > to 15.1.2.2 via ae1.0, Push 1131, Push 45103(top)
18 ...
19 42.0.49.0/24 *[BGP/170] 1d 17:24:52, MED 1, localpref 100, from 12.1.1.3
20     AS path: I, validation-state: unverified
21     > to 15.1.1.2 via ae0.0, Push 1131, Push 45103(top)
22     > to 15.1.2.2 via ae1.0, Push 1131, Push 45103(top)
23 42.0.50.0/24 *[BGP/170] 1d 17:24:52, MED 1, localpref 100, from 12.1.1.3
24     AS path: I, validation-state: unverified
25     > to 15.1.1.2 via ae0.0, Push 1131, Push 45103(top)
26     > to 15.1.2.2 via ae1.0, Push 1131, Push 45103(top)
27
28 regress@PE1>
```

1.3.6 CE-PE OSPFv3 Verification

OSPFv3 Neighborhood of the routing-instance is as shown below:

```
1 regress@PE1> show ospf3 neighbor instance METRO_VRF_2001
2 ID Interface State Pri Dead
3 209.0.0.1 et-0/0/2.1 Full 0 38
4   Neighbor-address fe80::228:1ff:fe00:1
5
6 regress@PE1>
```

The OSPFv3 routes received from the CE can be listed as

```
1 regress@PE1> show route table METRO_VRF_2001.inet6.0 protocol ospf3
2
3 METRO_VRF_2001.inet6.0: 107 destinations, 107 routes (107 active, 0 holddown, 0 hidden)
4 + = Active Route, - = Last Active, * = Both
5
6 3000:7:1:1::/64 *[OSPF3/10] 1d 17:28:54, metric 1
7     > to fe80::228:1ff:fe00:1 via et-0/0/2.1
8 3000:7:1:2::/64 *[OSPF3/10] 1d 17:28:54, metric 1
9     > to fe80::228:1ff:fe00:1 via et-0/0/2.1
10 ...
11 3000:7:1:31::/64 *[OSPF3/10] 1d 17:28:54, metric 1
12     > to fe80::228:1ff:fe00:1 via et-0/0/2.1
13 3000:7:1:32::/64 *[OSPF3/10] 1d 17:28:54, metric 1
14     > to fe80::228:1ff:fe00:1 via et-0/0/2.1
15 ff02::5/128 *[OSPF3/10] 1d 18:19:22, metric 1
16     MultiRecv
17 ff02::6/128 *[OSPF3/10] 1d 18:19:22, metric 1
18     MultiRecv
19
20 regress@PE1>
```

The OSPFv3 received from the PE2 routes as BGP routes over the 6VPE BGP family can be listed as follows.

```
1 regress@PE1> show route table METRO_VRF_2001.inet6.0 protocol bgp
2
3 METRO_VRF_2001.inet6.0: 107 destinations, 107 routes (107 active, 0 holddown, 0 hidden)
4 + = Active Route, - = Last Active, * = Both
```

```

5
6 2002:17:10:1::/64 *[BGP/170] 1d 18:07:53, localpref 100, from 12.1.1.3
7     AS path: I, validation-state: unverified
8     > to 15.1.1.2 via ae0.0, Push 1131, Push 2, Push 45103(top)
9     to 15.1.2.2 via ae1.0, Push 1131, Push 2, Push 45103(top)
10 3000:6:1:1::/64 *[BGP/170] 1d 17:30:42, MED 1, localpref 100, from 12.1.1.3
11     AS path: I, validation-state: unverified
12     > to 15.1.1.2 via ae0.0, Push 1131, Push 2, Push 45103(top)
13     to 15.1.2.2 via ae1.0, Push 1131, Push 2, Push 45103(top)
14 ...
15 3000:6:1:31::/64 *[BGP/170] 1d 17:30:42, MED 1, localpref 100, from 12.1.1.3
16     AS path: I, validation-state: unverified
17     > to 15.1.1.2 via ae0.0, Push 1131, Push 2, Push 45103(top)
18     to 15.1.2.2 via ae1.0, Push 1131, Push 2, Push 45103(top)
19 3000:6:1:32::/64 *[BGP/170] 1d 17:30:42, MED 1, localpref 100, from 12.1.1.3
20     AS path: I, validation-state: unverified
21     > to 15.1.1.2 via ae0.0, Push 1131, Push 2, Push 45103(top)
22     to 15.1.2.2 via ae1.0, Push 1131, Push 2, Push 45103(top)
23
24 regress@PE1>

```

1.3.7 CE-PE ISIS Verification

ISIS Adjacency of the routing-instance is as shown below:

```

1 regress@PE1> show isis adjacency instance METRO_VRF_3001
2 Interface System L State Hold (secs) SNPA
3 et-0/0/3.1 6401.0001.0000 2 Up 29
4
5 regress@PE1>

```

The ISIS routes received from the CE:

```

1 regress@PE1> show route table METRO_VRF_3001.inet.0 protocol isis
2
3 METRO_VRF_3001.inet.0: 105 destinations, 105 routes (105 active, 0 holddown, 0 hidden)
4 + = Active Route, - = Last Active, * = Both
5
6 44.0.1.0/24 *[IS-IS/18] 1d 18:10:17, metric 10
7     > to 16.14.1.2 via et-0/0/3.1
8 44.0.2.0/24 *[IS-IS/18] 1d 18:10:17, metric 10
9     > to 16.14.1.2 via et-0/0/3.1
10 ...
11 44.0.49.0/24 *[IS-IS/18] 1d 18:10:17, metric 10
12     > to 16.14.1.2 via et-0/0/3.1
13 44.0.50.0/24 *[IS-IS/18] 1d 18:10:17, metric 10
14     > to 16.14.1.2 via et-0/0/3.1
15
16 regress@PE1>

```

The ISIS routes received from the PE2 as BGP routes over the L3VPN BGP family can be listed as follows.

```

1 regress@PE1> show route table METRO_VRF_3001.inet.0 protocol bgp
2
3 METRO_VRF_3001.inet.0: 105 destinations, 105 routes (105 active, 0 holddown, 0 hidden)
4 + = Active Route, - = Last Active, * = Both
5
6 12.4.0.1/32 *[BGP/170] 1d 18:16:45, localpref 100, from 12.1.1.3
7     AS path: I, validation-state: unverified
8     to 15.1.1.2 via ae0.0, Push 2242, Push 45103(top)
9     > to 15.1.2.2 via ae1.0, Push 2242, Push 45103(top)
10 17.14.1.0/24 *[BGP/170] 1d 18:16:45, localpref 100, from 12.1.1.3
11     AS path: I, validation-state: unverified
12     to 15.1.1.2 via ae0.0, Push 2242, Push 45103(top)
13     > to 15.1.2.2 via ae1.0, Push 2242, Push 45103(top)
14 45.0.1.0/24 *[BGP/170] 1d 18:14:58, MED 10, localpref 100, from 12.1.1.3
15     AS path: I, validation-state: unverified
16     to 15.1.1.2 via ae0.0, Push 2242, Push 45103(top)
17     > to 15.1.2.2 via ae1.0, Push 2242, Push 45103(top)
18 45.0.2.0/24 *[BGP/170] 1d 18:14:58, MED 10, localpref 100, from 12.1.1.3
19     AS path: I, validation-state: unverified
20     to 15.1.1.2 via ae0.0, Push 2242, Push 45103(top)
21     > to 15.1.2.2 via ae1.0, Push 2242, Push 45103(top)
22 ...

```

```

23 45.0.49.0/24 *[BGP/170] 1d 18:14:58, MED 10, localpref 100, from 12.1.1.3
24     AS path: I, validation-state: unverified
25     > to 15.1.1.2 via ae0.0, Push 2242, Push 45103(top)
26     > to 15.1.2.2 via ae1.0, Push 2242, Push 45103(top)
27 45.0.50.0/24 *[BGP/170] 1d 18:14:58, MED 10, localpref 100, from 12.1.1.3
28     AS path: I, validation-state: unverified
29     > to 15.1.1.2 via ae0.0, Push 2242, Push 45103(top)
30     > to 15.1.2.2 via ae1.0, Push 2242, Push 45103(top)
31
32 regress@PE1>

```

1.3.8 CE-PE ISISv6 Verification

ISISv6 Adjacency of the routing-instance is as shown below:

```

1 regress@PE1> show isis adjacency instance METRO_VRF_3001
2 Interface System L State Hold (secs) SNPA
3 et-0/0/3.1 6401.0001.0000 2 Up 29
4
5 regress@PE1>

```

The ISISv6 routes received from the CE can be listed as

```

1 regress@PE1> show route table METRO_VRF_3001.inet6.0 protocol isis
2
3 METRO_VRF_3001.inet6.0: 28 destinations, 28 routes (28 active, 0 holddown, 0 hidden)
4 + = Active Route, - = Last Active, * = Both
5
6 3000:4:1:1::/64 *[IS-IS/18] 1d 18:17:26, metric 10
7     > to fe80::215:1ff:fe00:1 via et-0/0/3.1
8 3000:4:1:2::/64 *[IS-IS/18] 1d 18:17:26, metric 10
9     > to fe80::215:1ff:fe00:1 via et-0/0/3.1
10 ...
11 3000:4:1:9::/64 *[IS-IS/18] 1d 18:17:43, metric 10
12     > to fe80::215:1ff:fe00:1 via et-0/0/3.1
13 3000:4:1:a::/64 *[IS-IS/18] 1d 18:17:43, metric 10
14     > to fe80::215:1ff:fe00:1 via et-0/0/3.1
15
16 regress@PE1>

```

The ISISv6 routes received from the PE2 as BGP routes over the 6vpe bgp family can be listed as follows.

```

1 regress@PE1> show route table METRO_VRF_3001.inet6.0 protocol bgp
2
3 METRO_VRF_3001.inet6.0: 28 destinations, 28 routes (28 active, 0 holddown, 0 hidden)
4 + = Active Route, - = Last Active, * = Both
5
6 2002:17:14:1::/64 *[BGP/170] 1d 18:20:01, localpref 100, from 12.1.1.3
7     AS path: I, validation-state: unverified
8     > to 15.1.1.2 via ae0.0, Push 2242, Push 2, Push 45103(top)
9     > to 15.1.2.2 via ae1.0, Push 2242, Push 2, Push 45103(top)
10 2005:12:4::1/128 *[BGP/170] 1d 18:20:01, localpref 100, from 12.1.1.3
11     AS path: I, validation-state: unverified
12     > to 15.1.1.2 via ae0.0, Push 2242, Push 2, Push 45103(top)
13     > to 15.1.2.2 via ae1.0, Push 2242, Push 2, Push 45103(top)
14 3001:1:1:1::/64 *[BGP/170] 1d 18:18:18, MED 10, localpref 100, from 12.1.1.3
15     AS path: I, validation-state: unverified
16     > to 15.1.1.2 via ae0.0, Push 2242, Push 2, Push 45103(top)
17     > to 15.1.2.2 via ae1.0, Push 2242, Push 2, Push 45103(top)
18 3001:1:1:2::/64 *[BGP/170] 1d 18:18:18, MED 10, localpref 100, from 12.1.1.3
19     AS path: I, validation-state: unverified
20     > to 15.1.1.2 via ae0.0, Push 2242, Push 2, Push 45103(top)
21     > to 15.1.2.2 via ae1.0, Push 2242, Push 2, Push 45103(top)
22 ...
23 3001:1:1:9::/64 *[BGP/170] 1d 18:18:18, MED 10, localpref 100, from 12.1.1.3
24     AS path: I, validation-state: unverified
25     > to 15.1.1.2 via ae0.0, Push 2242, Push 2, Push 45103(top)
26     > to 15.1.2.2 via ae1.0, Push 2242, Push 2, Push 45103(top)
27 3001:1:1:a::/64 *[BGP/170] 1d 18:18:18, MED 10, localpref 100, from 12.1.1.3
28     AS path: I, validation-state: unverified
29     > to 15.1.1.2 via ae0.0, Push 2242, Push 2, Push 45103(top)
30     > to 15.1.2.2 via ae1.0, Push 2242, Push 2, Push 45103(top)
31
32 regress@PE1>

```

1.3.9 CE-PE Static Verification

The Static routes configured for the CE can be listed as

```
1 regress@PE1> show route table METRO_VRF_3501.inet.0 protocol static
2
3 METRO_VRF_3501.inet.0: 5 destinations, 5 routes (5 active, 0 holddown, 0 hidden)
4 + = Active Route, - = Last Active, * = Both
5
6 46.0.1.0/24 *[Static/5] 1d 18:33:39
7     > to 16.16.1.2 via et-0/0/4.1
8
9 regress@PE1>
```

The Static routes received from the PE2 as BGP routes over the L3VPN BGP family can be listed as follows.

```
1 regress@PE1> show route table METRO_VRF_3501.inet.0 protocol bgp
2
3 METRO_VRF_3501.inet.0: 5 destinations, 5 routes (5 active, 0 holddown, 0 hidden)
4 + = Active Route, - = Last Active, * = Both
5
6 17.16.1.0/24 *[BGP/170] 1d 18:24:25, localpref 100, from 12.1.1.3
7     AS path: I, validation-state: unverified
8     > to 15.1.1.2 via ae0.0, Push 2797, Push 45103(top)
9     to 15.1.2.2 via ae1.0, Push 2797, Push 45103(top)
10 46.10.1.0/24 *[BGP/170] 1d 18:24:25, localpref 100, from 12.1.1.3
11     AS path: I, validation-state: unverified
12     > to 15.1.1.2 via ae0.0, Push 2797, Push 45103(top)
13     to 15.1.2.2 via ae1.0, Push 2797, Push 45103(top)
14
15 regress@PE1>
```

1.3.10 CE-PE Staticv6 Verification

The Static-v6 routes received from the CE can be listed as

```
1 regress@PE1> show route table METRO_VRF_3501.inet6.0 protocol static
2
3 METRO_VRF_3501.inet6.0: 7 destinations, 7 routes (7 active, 0 holddown, 0 hidden)
4 + = Active Route, - = Last Active, * = Both
5
6 2008:1:0:1::/64 *[Static/5] 1d 18:35:38
7     > to 2002:16:16:1::2 via et-0/0/4.1
8
9 regress@PE1>
```

The Static-v6 routes received from the PE2 as BGP routes over the L3VPN BGP family can be listed as follows.

```
1 regress@PE1> show route table METRO_VRF_3501.inet.0 protocol bgp
2
3 METRO_VRF_3501.inet.0: 5 destinations, 5 routes (5 active, 0 holddown, 0 hidden)
4 + = Active Route, - = Last Active, * = Both
5
6 17.16.1.0/24 *[BGP/170] 1d 18:26:14, localpref 100, from 12.1.1.3
7     AS path: I, validation-state: unverified
8     > to 15.1.1.2 via ae0.0, Push 2797, Push 45103(top)
9     to 15.1.2.2 via ae1.0, Push 2797, Push 45103(top)
10 46.10.1.0/24 *[BGP/170] 1d 18:26:14, localpref 100, from 12.1.1.3
11     AS path: I, validation-state: unverified
12     > to 15.1.1.2 via ae0.0, Push 2797, Push 45103(top)
13     to 15.1.2.2 via ae1.0, Push 2797, Push 45103(top)
14
15 regress@PE1>
```

1.3.11 6PE Verification

The neighborship of the BGP Global IPv6 Peer is listed below

```

1 regress@PE1> show bgp neighbor 2002:16:100:1::2
2 Peer: 2002:16:100:1::2+32777 AS 65001 Local: 2002:16:100:1::1+179 AS 65000
3   Group: EBGp Routing-Instance: master
4   Forwarding routing-instance: master
5   Type: External State: Established Flags: <Sync>
6   Last State: OpenConfirm Last Event: RecvKeepAlive
7   Last Error: None
8   Options: <AddressFamily PeerAS Refresh As Override>
9   Options: <GracefulShutdownRcv>
10  Address families configured: inet6-unicast
11  Holdtime: 90 Preference: 170
12  Graceful Shutdown Receiver local-preference: 0
13  Number of flaps: 0
14  Receive eBGP Origin Validation community: Reject
15  Peer ID: 201.0.0.1 Local ID: 12.1.1.1 Active Holdtime: 90
16  Keepalive Interval: 30 Group index: 1 Peer index: 0 SNMP index: 2
17  I/O Session Thread: bgpio-0 State: Enabled
18  BFD: disabled, down
19  Local Interface: et-0/0/1.4000
20  NLRI for restart configured on peer: inet6-unicast
21  NLRI advertised by peer: inet-unicast inet-multicast inet-vpn-unicast inet6-unicast
    multicast l2vpn inet6-vpn-unicast
22  NLRI for this session: inet6-unicast
23  Peer supports Refresh capability (2)
24  Stale routes from peer are kept for: 300
25  Peer does not support Restarter functionality
26  NLRI of all end-of-rib markers sent: inet6-unicast
27  Peer does not support Receiver functionality
28  Peer does not support LLGR Restarter or Receiver functionality
29  Peer does not support 4 byte AS extension
30  Peer does not support Addpath
31  NLRI that peer supports extended nexthop encoding for: inet-unicast inet-multicast inet-vpn-unicast
32  Table inet6.0 Bit: 30001
33  RIB State: BGP restart is complete
34  Send state: in sync
35  Active prefixes: 15000
36  Received prefixes: 15000
37  Accepted prefixes: 15000
38  Suppressed due to damping: 0
39  Advertised prefixes: 15002
40  Last traffic (seconds): Received 21 Sent 16 Checked 147443
41  Input messages: Total 4951 Updates 34 Refreshes 0 Octets 231303
42  Output messages: Total 4994 Updates 36 Refreshes 0 Octets 232009
43  Output Queue[2]: 0 (inet6.0, inet6-unicast)
44
45 regress@PE1>

```

The 6PE routes received from the Global Peer and Remote Peer PE2 is as shown below:

```

1 regress@PE1> show route table inet6.0
2
3 inet6.0: 30008 destinations, 30008 routes (30008 active, 0 holddown, 0 hidden)
4 + = Active Route, - = Last Active, * = Both
5
6 2002:12:1:1::1/128 *[Direct/0] 1d 17:10:57
7   > via lo0.0
8 2002:12:1:1::3/128 *[BGP/170] 1d 16:59:24, localpref 100, from 12.1.1.3
9   AS path: I, validation-state: unverified
10   > to 15.1.1.2 via ae0.0, Push 2, Push 45103(top)
11   to 15.1.2.2 via ae1.0, Push 2, Push 45103(top)
12 2002:16:100:1::/64 *[Direct/0] 1d 17:09:13
13   > via et-0/0/1.4000
14 2002:16:100:1::1/128
15   *[Local/0] 1d 17:09:13
16   Local via et-0/0/1.4000
17 2002:17:100:1::/64 *[BGP/170] 1d 16:59:24, localpref 100, from 12.1.1.3
18   AS path: I, validation-state: unverified
19   > to 15.1.1.2 via ae0.0, Push 2, Push 45103(top)
20   to 15.1.2.2 via ae1.0, Push 2, Push 45103(top)
21 2002:50:0:1::/64 *[BGP/170] 1d 16:58:26, localpref 100
22   AS path: 65001 I, validation-state: unverified
23   > to 2002:16:100:1::2 via et-0/0/1.4000
24 2002:50:0:2::/64 *[BGP/170] 1d 16:58:26, localpref 100
25   AS path: 65001 I, validation-state: unverified
26   > to 2002:16:100:1::2 via et-0/0/1.4000
27 2002:50:0:3::/64 *[BGP/170] 1d 16:58:26, localpref 100
28   AS path: 65001 I, validation-state: unverified
29   > to 2002:16:100:1::2 via et-0/0/1.4000

```

```

30 ...
31 2002:55:0:3a96::/64*[BGP/170] 1d 16:59:11, localpref 100, from 12.1.1.3
32     AS path: 65001 I, validation-state: unverified
33     to 15.1.1.2 via ae0.0, Push 4018, Push 2, Push 45103(top)
34     > to 15.1.2.2 via ae1.0, Push 4018, Push 2, Push 45103(top)
35 2002:55:0:3a97::/64*[BGP/170] 1d 16:59:11, localpref 100, from 12.1.1.3
36     AS path: 65001 I, validation-state: unverified
37     to 15.1.1.2 via ae0.0, Push 4018, Push 2, Push 45103(top)
38     > to 15.1.2.2 via ae1.0, Push 4018, Push 2, Push 45103(top)
39 2002:55:0:3a98::/64*[BGP/170] 1d 16:59:11, localpref 100, from 12.1.1.3
40     AS path: 65001 I, validation-state: unverified
41     > to 15.1.1.2 via ae0.0, Push 4018, Push 2, Push 45103(top)
42     to 15.1.2.2 via ae1.0, Push 4018, Push 2, Push 45103(top)
43 fe80::ae78:d10f:a02d:350c/128
44     *[Local/0] 1d 17:10:02
45     Local via et-0/0/1.4000
46 fe80::ae78:d1f0:2d:3503/128
47     *[Direct/0] 1d 17:11:44
48     > via lo0.0
49 ff02::2/128 *[INET6/0] 1d 17:12:42
50     MultiRecv
51
52 regress@PE1>

```

1.3.12 Global IPv4 Verification

The neighborhood of the BGP Global IPv4 Peer is listed below:

```

1 regress@PE1> show bgp neighbor 16.100.1.2
2 Peer: 16.100.1.2+32779 AS 65001 Local: 16.100.1.1+179 AS 65000
3 Group: EBGP Routing-Instance: master
4 Forwarding routing-instance: master
5 Type: External State: Established Flags: <Sync>
6 Last State: OpenConfirm Last Event: RecvKeepAlive
7 Last Error: None
8 Options: <AddressFamily PeerAS Refresh As Override>
9 Options: <GracefulShutdownRcv>
10 Address families configured: inet-unicast
11 Holdtime: 90 Preference: 170
12 Graceful Shutdown Receiver local-preference: 0
13 Number of flaps: 0
14 Receive eBGP Origin Validation community: Reject
15 Peer ID: 200.0.0.1 Local ID: 12.1.1.1 Active Holdtime: 90
16 Keepalive Interval: 30 Group index: 1 Peer index: 1 SNMP index: 1
17 I/O Session Thread: bgpio-0 State: Enabled
18 BFD: disabled, down
19 Local Interface: et-0/0/1.4000
20 NLRI for restart configured on peer: inet-unicast
21 NLRI advertised by peer: inet-unicast inet-multicast inet-vpn-unicast inet6-unicast inet6-
    multicast l2vpn inet6-vpn-unicast
22 NLRI for this session: inet-unicast
23 Peer supports Refresh capability (2)
24 Stale routes from peer are kept for: 300
25 Peer does not support Restarter functionality
26 NLRI of all end-of-rib markers sent: inet-unicast
27 Peer does not support Receiver functionality
28 Peer does not support LLGR Restarter or Receiver functionality
29 Peer does not support 4 byte AS extension
30 Peer does not support Addpath
31 Table inet.0 Bit: 20001
32 RIB State: BGP restart is complete
33 Send state: in sync
34 Active prefixes: 30000
35 Received prefixes: 30000
36 Accepted prefixes: 30000
37 Suppressed due to damping: 0
38 Advertised prefixes: 30002
39 Last traffic (seconds): Received 17 Sent 4 Checked 147619
40 Input messages: Total 4953 Updates 30 Refreshes 0 Octets 215047
41 Output messages: Total 4998 Updates 34 Refreshes 0 Octets 215788
42 Output Queue[1]: 0 (inet.0, inet-unicast)
43
44 regress@PE1>

```

The IPv4 routes received from the Global Peer and Remote Peer PE2 is as shown below:

```

1 regress@PE1> show route table inet.0 protocol bgp
2
3 inet.0: 60057 destinations, 60090 routes (60057 active, 0 holddown, 0 hidden)
4 + = Active Route, - = Last Active, * = Both
5
6 ...
7
8 50.0.1.0/24 *[BGP/170] 1d 19:15:32, localpref 100
9      AS path: 65001 I, validation-state: unverified
10     > to 16.100.1.2 via et-0/0/1.4000
11 50.0.2.0/24 *[BGP/170] 1d 19:15:32, localpref 100
12      AS path: 65001 I, validation-state: unverified
13     > to 16.100.1.2 via et-0/0/1.4000
14 ...
15
16 55.117.47.0/24 *[BGP/170] 1d 19:16:33, localpref 100, from 12.1.1.3
17      AS path: 65001 I, validation-state: unverified
18     to 15.1.1.2 via ae0.0, Push 45103
19     > to 15.1.2.2 via ae1.0, Push 45103
20 55.117.48.0/24 *[BGP/170] 1d 19:16:33, localpref 100, from 12.1.1.3
21      AS path: 65001 I, validation-state: unverified
22     > to 15.1.1.2 via ae0.0, Push 45103
23     to 15.1.2.2 via ae1.0, Push 45103
24 ...
25
26 regress@PE1>

```

1.4 Verifying Feature At Scale

The following output captures BGP, OSPF & OSPFv3 peering scale:

```

1 regress@PE1> show bgp summary |match Estab | count
2 Count: 2503 lines
3
4 regress@PE1> show ospf neighbor instance all |match Full | count
5 Count: 1002 lines
6
7 regress@PE1> show ospf3 neighbor instance all |match Full | count
8 Count: 100 lines

```

```

1 regress@PE1> show route summary
2 Autonomous system number: 65000
3 Router ID: 12.1.1.1
4
5 Highwater Mark (All time / Time averaged watermark)
6   RIB unique destination routes: 2032662 at 2022-10-26 21:14:00 / 2032660
7   RIB routes : 2032671 at 2022-10-26 21:14:00 / 2032669
8   FIB routes : 1377491 at 2022-10-25 04:57:29 / 1377491
9   VRF type routing instances : 4000 at 2022-10-25 03:35:04
10
11 inet.0: 60031 destinations, 60038 routes (60031 active, 0 holddown, 0 hidden)
12   Direct: 5 routes, 5 active
13   Local: 4 routes, 4 active
14   OSPF: 6 routes, 6 active
15   BGP: 60009 routes, 60002 active
16   Static: 14 routes, 14 active
17
18 inet.3: 2 destinations, 2 routes (2 active, 0 holddown, 0 hidden)
19   L-OSPF: 2 routes, 2 active
20
21 METRO_VRF_1.inet.0: 503 destinations, 503 routes (503 active, 0 holddown, 0 hidden)
22   Direct: 1 routes, 1 active
23   Local: 1 routes, 1 active
24   BGP: 501 routes, 501 active
25
26 METRO_VRF_2.inet.0: 503 destinations, 503 routes (503 active, 0 holddown, 0 hidden)
27   Direct: 1 routes, 1 active
28   Local: 1 routes, 1 active
29   BGP: 501 routes, 501 active
30 ...
31 METRO_VRF_2001.inet.0: 105 destinations, 105 routes (105 active, 0 holddown, 0 hidden)
32   Direct: 1 routes, 1 active
33   Local: 1 routes, 1 active
34   OSPF: 52 routes, 52 active
35   BGP: 51 routes, 51 active

```

```

36 ...
37 METRO_VRF_3001.inet.0: 105 destinations, 105 routes (105 active, 0 holddown, 0 hidden)
38     Direct: 2 routes, 2 active
39     Local: 1 routes, 1 active
40     BGP: 52 routes, 52 active
41     IS-IS: 50 routes, 50 active
42 ...
43 METRO_VRF_3501.inet.0: 5 destinations, 5 routes (5 active, 0 holddown, 0 hidden)
44     Direct: 1 routes, 1 active
45     Local: 1 routes, 1 active
46     BGP: 2 routes, 2 active
47     Static: 1 routes, 1 active
48 ...
49 mpls.0: 4014 destinations, 4014 routes (4014 active, 0 holddown, 0 hidden)
50     MPLS: 6 routes, 6 active
51     VPN: 4001 routes, 4001 active
52     L-OSPF: 7 routes, 7 active
53
54 bgp.l3vpn.0: 580000 destinations, 580000 routes (580000 active, 0 holddown, 0 hidden)
55     BGP: 580000 routes, 580000 active
56
57 inet6.0: 30008 destinations, 30008 routes (30008 active, 0 holddown, 0 hidden)
58     Direct: 3 routes, 3 active
59     Local: 2 routes, 2 active
60     BGP: 30002 routes, 30002 active
61     INET6: 1 routes, 1 active
62
63 inet6.3: 2 destinations, 2 routes (2 active, 0 holddown, 0 hidden)
64     L-OSPF: 2 routes, 2 active
65
66 METRO_VRF_1.inet6.0: 205 destinations, 205 routes (205 active, 0 holddown, 0 hidden)
67     Direct: 1 routes, 1 active
68     Local: 2 routes, 2 active
69     BGP: 201 routes, 201 active
70     INET6: 1 routes, 1 active
71 ...
72 METRO_VRF_2001.inet6.0: 107 destinations, 107 routes (107 active, 0 holddown, 0 hidden)
73     Direct: 1 routes, 1 active
74     Local: 2 routes, 2 active
75     OSPF3: 52 routes, 52 active
76     BGP: 51 routes, 51 active
77     INET6: 1 routes, 1 active
78 ...
79 METRO_VRF_3001.inet6.0: 28 destinations, 28 routes (28 active, 0 holddown, 0 hidden)
80     Direct: 3 routes, 3 active
81     Local: 2 routes, 2 active
82     BGP: 12 routes, 12 active
83     IS-IS: 10 routes, 10 active
84     INET6: 1 routes, 1 active
85 ...
86 METRO_VRF_3501.inet6.0: 7 destinations, 7 routes (7 active, 0 holddown, 0 hidden)
87     Direct: 1 routes, 1 active
88     Local: 2 routes, 2 active
89     BGP: 2 routes, 2 active
90     Static: 1 routes, 1 active
91     INET6: 1 routes, 1 active
92 ...
93 bgp.l3vpn-inet6.0: 61700 destinations, 61700 routes (61700 active, 0 holddown, 0 hidden)
94     BGP: 61700 routes, 61700 active
95
96 regress@PE1>

```

1.4.1 Traffic Verification at Router at PE1

Traffic from the CE flows towards the core over the two LAG bundles ae0 and ae1 and traffic load-balanced across the bundles and the member interfaces of the bundle.

```

1 PE1 Seconds: 6 Time: 20:32:01
2
3 Interface Link Input packets (pps) Output packets (pps)
4
5 ae0 Up 5586190244536 (38943485) 5575357824202 (38871357)
6 ae1 Up 5555772552639 (38763433) 5566579800978 (38841649)
7 esi Up 0 (0) 0 (0)
8 et-0/0/0 Up 2228214479834 (15511380) 2228214835244 (15513576)
9 et-0/0/1 Up 2229235576443 (15520844) 2229235532415 (15521822)

```

```

10 et-0/0/2 Up 2228332663250 (15517930) 2227956502538 (15518860)
11 et-0/0/3 Up 2228318956042 (15518420) 2228336007745 (15519712)
12 et-0/0/4 Up 2228317159060 (15523076) 2228317066058 (15524683)
13 et-0/0/5 Down 0 (0) 0 (0)
14 et-0/0/6 Up 1410137465761 (9827780) 1390248083194 (9689916)
15 et-0/0/7 Up 1405078195401 (9794287) 1397493349083 (9742324)
16 et-0/0/8 Up 1369119218901 (9545572) 1392702964826 (9710874)
17 et-0/0/9 Up 1401855364473 (9775846) 1394913427099 (9728243)
18 et-0/0/10 Up 1392259351671 (9710934) 1391134693656 (9703807)
19 et-0/0/11 Up 1389924844295 (9696679) 1390350321922 (9700313)
20 et-0/0/12 Up 1381150822629 (9637431) 1395562427879 (9738708)
21 et-0/0/13 Up 1392437534044 (9718389) 1389532357521 (9698821)

```

1.5 Generating Scale Configuration

The Python Script is having three components, a Jinja File capturing the various configuration templates, a Params File capturing the various user variables and the script uses both these input files and generates the required configurations and commit that in the router.

Note: The configs are generated as per the CE-bound interface schema.

1.5.1 Jinja Configuration Template

The template contains configurations for both interfaces which are part of the routing instances and routing instance configurations.

```

1  # Routing Instnace Configuration
2  # File Name: l3vpn_jinja.j2
3  # Version: 1.0
4
5  {%- if ce_pe_is_bgpv4 %}
6  groups {{group_name}} {
7      routing-instances {
8          {%- for vrf_id, ifl_name, peeras, neighborip4 in l3vpn_variables %}
9          METRO_VRF_{{vrf_id}} {
10             instance-type vrf;
11             interface {{ifl_name}};
12             route-distinguisher {{router_id}}:{{vrf_id}};
13             vrf-target target:{{local_as_no}}:{{vrf_id}};
14             vrf-table-label;
15             protocols {
16                 bgp {
17                     group METRO_VRF_EBGpV4_{{vrf_id}} {
18                         type external;
19                         family inet {
20                             unicast {
21
22                             }
23                         }
24                         peer-as {{peeras}};
25                         as-override;
26                         neighbor {{neighborip4}};
27                     }
28                 }
29             }
30         }
31     }
32 }
33 {%- endif %}
34 {%- if ce_pe_is_bgpv6 %}
35 groups {{group_name}} {
36     routing-instances {
37         {%- for vrf_id, ifl_name, peeras, neighborip6 in vpnv6_variables %}
38         METRO_VRF_{{vrf_id}} {
39             instance-type vrf;
40             interface {{ifl_name}};
41             route-distinguisher {{router_id}}:{{vrf_id}};
42             vrf-target target:{{local_as_no}}:{{vrf_id}};
43             vrf-table-label;
44             protocols {

```

```

45         bgp {
46             group METRO_VRF_EBGPv6_{{vrf_id}} {
47                 type external;
48                 family inet6 {
49                     unicast {
50                         }
51                     }
52                 peer-as {{peeras}};
53                 as-override;
54                 neighbor {{neighborip6}};
55             }
56         }
57     }
58 }
59 {%- endfor %}
60 }
61 }
62 {%- endif %}
63 {%- if ce_pe_is_ospf %}
64 groups {{group_name}} {
65     routing-instances {
66         {%- for vrf_id, ifl_name in ospf_variables %}
67         METRO_VRF_{{vrf_id}} {
68             instance-type vrf;
69             interface {{ifl_name}};
70             route-distinguisher {{router_id}}:{{vrf_id}};
71             vrf-target target:{{local_as_no}}:{{vrf_id}};
72             vrf-table-label;
73             protocols {
74                 ospf {
75                     area 0.0.0.0 {
76                         interface {{ifl_name}} {
77                             interface-type p2p;
78                         }
79                     }
80                 }
81             }
82         }
83         {%- endfor %}
84     }
85 }
86 {%- endif %}
87 {%- if ce_pe_is_ospfv6 %}
88 groups {{group_name}} {
89     routing-instances {
90         {%- for vrf_id, ifl_name in ospf_variables %}
91         METRO_VRF_{{vrf_id}} {
92             instance-type vrf;
93             interface {{ifl_name}};
94             route-distinguisher {{router_id}}:{{vrf_id}};
95             vrf-target target:{{local_as_no}}:{{vrf_id}};
96             vrf-table-label;
97             protocols {
98                 ospf3 {
99                     area 0.0.0.0 {
100                         interface {{ifl_name}} {
101                             interface-type p2p;
102                         }
103                     }
104                 }
105             }
106         }
107         {%- endfor %}
108     }
109 }
110 {%- endif %}
111 {%- if ce_pe_is_isis %}
112 groups {{group_name}} {
113     interfaces {
114         lo0 {
115             {%- for unit_id, ip_addr, iso_addr in lo0_variables %}
116             unit {{unit_id}} {
117                 family inet {
118                     address {{ip_addr}};
119                 }
120                 family iso {
121                     address {{iso_addr}};
122                 }
123             }
124             {%- endfor %}
125         }

```

```

126     }
127     routing-instances {
128         {%- for vrf_id, ifl_name, lo0_ifl in isis_variables %}
129         METRO_VRF_{{vrf_id}} {
130             instance-type vrf;
131             interface {{ifl_name}};
132             interface {{lo0_ifl}};
133             route-distinguisher {{router_id}}:{{vrf_id}};
134             vrf-target target:{{local_as_no}}:{{vrf_id}};
135             vrf-table-label;
136             protocols {
137                 isis {
138                     interface {{ifl_name}} {
139                         level 1 disable;
140                         point-to-point;
141                     }
142                     interface {{lo0_ifl}} {
143                         passive;
144                     }
145                 }
146             }
147         }
148         {%- endfor %}
149     }
150 }
151 {%- endif %}
152 {%- if ce_pe_is_isisv6 %}
153 groups {{group_name}} {
154     interfaces {
155         lo0 {
156             {%- for unit_id, ipv6_addr, iso_addr in lo0v6_variables %}
157             unit {{unit_id}} {
158                 family inet6 {
159                     address {{ipv6_addr}};
160                 }
161                 family iso {
162                     address {{iso_addr}};
163                 }
164             }
165             {%- endfor %}
166         }
167     }
168     routing-instances {
169         {%- for vrf_id, ifl_name, lo0_ifl in isisv6_variables %}
170         METRO_VRF_{{vrf_id}} {
171             instance-type vrf;
172             interface {{ifl_name}};
173             interface {{lo0_ifl}};
174             route-distinguisher {{router_id}}:{{vrf_id}};
175             vrf-target target:{{local_as_no}}:{{vrf_id}};
176             vrf-table-label;
177             protocols {
178                 isis {
179                     interface {{ifl_name}} {
180                         level 1 disable;
181                         point-to-point;
182                     }
183                     interface {{lo0_ifl}} {
184                         passive;
185                     }
186                 }
187             }
188         }
189         {%- endfor %}
190     }
191 }
192 {%- endif %}
193 {%- if ce_pe_is_static %}
194 groups {{group_name}} {
195     routing-instances {
196         {%- for vrf_id, ifl_name, static_route, next_hop_value in static_variables %}
197         METRO_VRF_{{vrf_id}} {
198             instance-type vrf;
199             interface {{ifl_name}};
200             route-distinguisher {{router_id}}:{{vrf_id}};
201             vrf-target target:{{local_as_no}}:{{vrf_id}};
202             vrf-table-label;
203             routing-options {
204                 static {
205                     route {{static_route}} next-hop {{next_hop_value}};
206                 }

```

```
207     }
208   }
209   {%~ endfor %}
210 }
211 }
212 {%~ endif %}
213 {%~ if ce_pe_is_staticv6 %}
214 groups {{group_name}} {
215   routing-instances {
216     {%~ for vrf_id, ifl_name, static_routev6, next_hop_valuev6 in staticv6_variables %}
217     METRO_VRF_{{vrf_id}} {
218       instance-type vrf;
219       interface {{ifl_name}};
220       route-distinguisher {{router_id}}:{{vrf_id}};
221       vrf-target target:{{local_as_no}}:{{vrf_id}};
222       vrf-table-label;
223       routing-options {
224         rib METRO_VRF_{{vrf_id}}.inet6.0 {
225           static {
226             route {{static_routev6}} next-hop {{next_hop_valuev6}};
227           }
228         }
229       }
230     }
231     {%~ endfor %}
232   }
233 }
234 {%~ endif %}
```

1.5.2 Configuration Parameters

““



DC – AI – Routing – Switching – Security – Wireless – Automation

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